

Responsive SVG Art Direction via @container Queries

Injecting responsive logic directly into vector graphics for infinite, DOM-independent scalability.

1. Self-Aware Canvas

The `inline-size` property turns the SVG itself into a queryable container context, completely independent of the DOM parent constraints.

2. Internal Logic

CSS `@container` queries read the SVG's measured width and selectively toggle internal layers, scales, and strokes dynamically.

3. Single Source of Truth

One unified vector file contains all possible visual states simultaneously, eliminating duplicate image assets and framework conditionals.

```
chart-component.svg

<svg class="responsive-graph" style="container-type: inline-size;">
  <style>

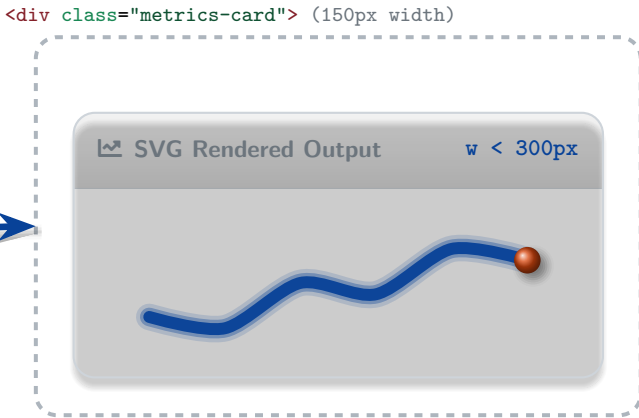
    /* --- BASE: Minimal Icon (Width < 300px) --- */
    .grid, .axes, .labels, .data-nodes, .legend { display: none; }
    .trend { stroke-width: 12px; stroke: #89b4fa; }
    .area-fill { display: none; }

    /* --- MEDIUM: Sidebar Chart (Width >= 300px) --- */
    @container (min-width: 300px) {
      .axes { display: block; stroke: #6c7086; }
      .trend { stroke-width: 6px; }
      .area-fill { display: block; opacity: 0.2; }
    }

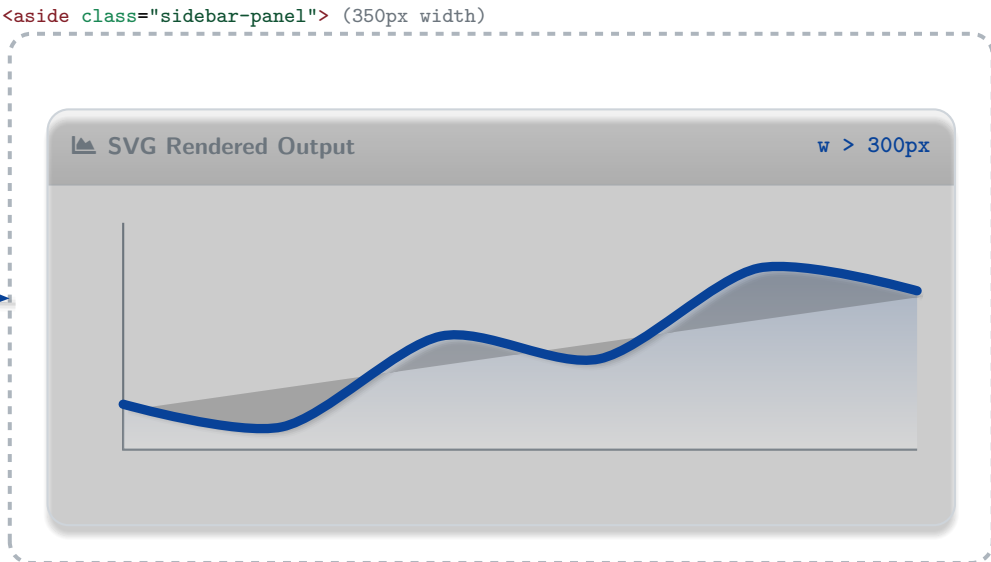
    /* --- LARGE: Data Dashboard (Width >= 600px) --- */
    @container (min-width: 600px) {
      .grid { display: block; }
      .labels { display: block; }
      .data-nodes { display: block; }
      .legend { display: block; }
      .axes { stroke: #a6adc8; }
    }
  </style>

  <!-- SVG DOM Structure -->
  <g class="grid"> <!-- 10x Grid Lines --> </g>
  <g class="axes"> <!-- X & Y Axis paths --> </g>
  <g class="labels"> <!-- Axis labels --> </g>
  <path class="area-fill" d="M10,50 L20,30..." />
  <path class="trend" d="M10,50 L20,30..." />
  <g class="data-nodes"> <!-- Data point circles --> </g>
  <g class="legend"> <!-- Text & markers --> </g>
</svg>
```

Applies to Icon



Applies to Sidebar



Applies to Dashboard

